

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0$ (1) • $= 44.8 \text{ t ha}^{-1} \text{ yr}^{-1}$ (accept 44.7-44.8) (1)	2
01.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25$ (1) • $= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}$ (1)	2
01.3	<i>1 mark for each point.</i> • Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall (1) • Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable (1) • Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land ($C = 0.02$) needed to bring loss below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ (1)	3
01.4	<i>1 mark for each point.</i> • Factors are averages / arbitrary values for typical conditions; rainfall erosivity varies between years / storms (1) • Does not include gully erosion, landslides, wind erosion or local variation in soil; based on data from particular regions (1)	2
01.5	<i>1 mark.</i> • C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	

Question	Marking guidance	Marks
02.1	1 mark for each point. - Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s (1) - Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline (1)	2
02.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. $(225 - 95) / 225 \times 100$ (1) $= 57.8\%$ (accept 58%) (1)	2
02.3	1 mark for each point. - Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form (1) - Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter (1) - The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated (1) - When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone; levels recover by December as the vortex breaks down (1)	4
02.4	1 mark for each point. - Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation (1) - Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
03.1	1 mark for each point. - Bacteria in the gut convert nitrates to nitrites (1) - Nitrites react with haemoglobin so it cannot carry oxygen (1) - Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster; usually from shallow wells receiving arable drainage (1)	3
03.2	1 mark for each point. - Nitrites may be converted in the gut to nitrosamines which are carcinogenic (1) - But most research has been on rats / the evidence in humans is not conclusive; nitrites are also ingested in processed food so the source cannot be isolated (1)	2
03.3	Any 3 from the points below - 1 mark each, maximum 3. - Reduce use of nitrate fertilisers / match application to crop need (1) - Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) (1) - Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones (1) - Store manure / slurry where it cannot leach or wash into rivers (1) - Grow crops with lower nitrogen requirements (eg triticale instead of wheat) (1) - Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser (1)	3

Question	Marking guidance	Marks
03.4	<i>1 mark for each point.</i> • Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) (1) • Lakes: dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
04.1	<i>1 mark for each point.</i> • Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions (1) • Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected; recirculation increases the metal concentration (1) • Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) (1) • Smelting needs chemical reduction at high temperatures; energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic (1) • Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes (1)	5
04.2	<i>1 mark for each point.</i> • Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves (1) • The vegetation is harvested and incinerated (1) • The concentrated metal in the ash is dissolved in acid and separated by electrolysis; also decontaminates polluted sites (1)	3
04.3	<i>1 mark for each point.</i> • Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) (1) • The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
05.1	<i>1 mark for each point.</i> • Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant (1) • Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate (1) • Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers (1) • Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor (1) • Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away (1) • Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions (1)	6

Question	Marking guidance	Marks
05.2	<i>1 mark for each point.</i> - The organisation / person that causes the pollution is responsible for (paying for) the damage caused (1) - If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely (1)	2
05.3	<i>1 mark for each point.</i> - It is assumed that the waste will cause pollution / harm until research shows it is unlikely to (1) - So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $(60 \times 45) / 9$ (1) $= 300$ (1)	2
06.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Marked individuals mix freely / randomly with the rest of the population (1) - No births, deaths, immigration or emigration between samples (1) - Marking does not affect survival or behaviour (eg make them more visible to predators) (1) - Marks do not wear off / fall off (1) - All individuals have the same chance of being caught (traps do not attract or repel marked animals) (1)	4
06.3	<i>1 mark for each point.</i> - Fewer marked beetles would survive to be recaptured (1) - So the proportion marked in the second sample would be too low and the population would be overestimated (1)	2
06.4	<i>1 mark for each point.</i> - Enough time for marked beetles to mix with the population (1) - Not so long that births / deaths / migration or loss of marks change the population (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
07.1	<i>1 mark for each point.</i> - Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen (1) - It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury (1)	2
07.2	<i>1 mark for each point.</i> - Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted (1) - So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) (1) - It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata) (1)	3

Question	Marking guidance	Marks
07.3	<i>1 mark for each point.</i> - Mercury is persistent / not degraded; it is an element so cannot be broken down (1) - It remains in the sediments and continues to be released / converted and taken up by the food chain; long-lived fish continue to accumulate it (1)	2
07.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Water current velocity and direction (dispersal) (1) - Properties of the pollutant: solubility, persistence, liposolubility / potential to bioaccumulate (1) - Food chain pathways to humans (fish, shellfish) / the critical group (1) - Sediment type / adsorption onto clay (1) - Volume of the water body / residence time / dilution (1) - Sensitive locations downstream (fisheries, drinking water abstraction) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
08.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $(15.0 - 2.2) / 15.0 \times 100$ (1) $= 85.3\%$ (accept 85%) (1)	2
08.2	<i>1 mark for each point.</i> - Added as an anti-knock agent to smooth combustion and reduce engine wear (1) - Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food (1) - Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children (1)	3
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate (1) - Phasing out of lead-based paint / safer paint removal (1) - Lead-free electrical solder (1) - Restrictions on lead fishing weights / shotgun pellets (1) - Industrial emission controls (1)	2
08.4	<i>1 mark for each point.</i> - Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe (1) - This prevents lead dissolving into the water (1)	2
08.5	<i>1 mark.</i> Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	
09.1	<i>Correct answer only.</i> B - Low liposolubility and low persistence (1)	1

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown (1) • Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain (1)	2
09.3	<i>1 mark for each point.</i> • They are intended to be toxic (to the target species) (1) • They are deliberately released / spread into the environment (rather than being an accidental or waste release) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted • Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites • Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G • Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use • Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement	9
10.2	<i>1 mark.</i> • Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25